

fareBeep: Use Case Specification

Target: Universal Nigerian Domestic Travel

1. Scenario Goal

Enable a user to identify a price drop and complete a secure booking on WhatsApp within a 10-minute guaranteed price window.

2. The Shared Ledger (Efficiency)

User A searches LOS-ABV. The result is cached in Supabase. User B (and 500 others) receives that same data instantly. This reduces Tiqwa API costs by 90% and ensures sub-second response times.

3. The 'Beep' (Proactive Alerting)

The system monitors the Ledger. When the price for a tracked route drops by > 10%, a 'Utility' message is sent via Meta Cloud API. This high-intent alert minimizes messaging costs while maximizing conversion.

4. The 10-Minute Handshake (Financial Safety)

To protect against Nigeria's volatile seat buckets, the moment a user initiates booking, a 10-minute timer starts. This 'Price Lock' is the handshake between the cached Ledger and the Live Inventory.

5. Settlement & Fulfillment

Payment is handled via Paystack. A 2-minute 'Bank Grace Period' is added to the timer to account for NIBSS network latency. Upon successful webhook verification, the ticket is issued automatically.

6. Flight Status (The Value-Add)

Post-booking, fareBeep monitors the physical aircraft location via Aviationstack. The user receives a 'Status Beep' if the plane is delayed, often before the airline gate announces it.